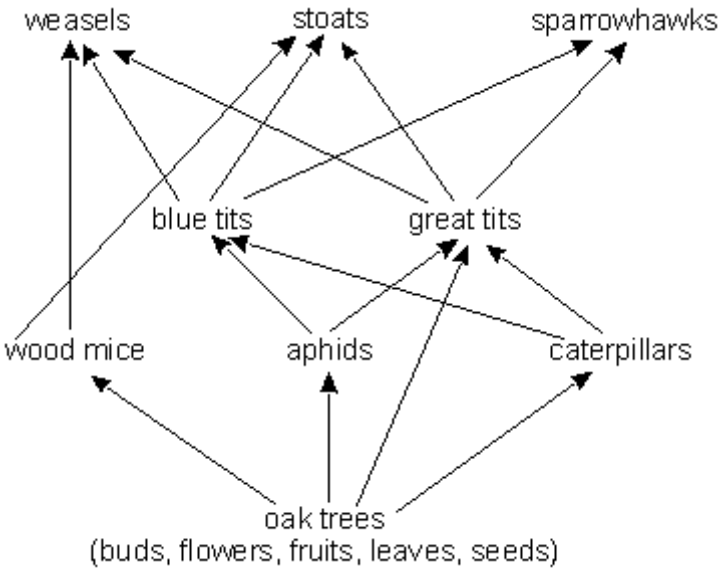
Q1. The diagram shows part of a food web in a wood.



Use **only** the information in the diagram to answer questions (a) and (b).

(a) Name **one** herbivore and **one** omnivore.

herbivore .....

omnivore .....

2 marks

(b) The number of blue tits in the wood decreases.
This affects the number of great tits in the wood.

(i) Give **one** reason why the number of great tits might **increase**.

.....

.....

1 mark

(ii) Give **one** reason why the number of great tits might **decrease**.

.....

.....

1 mark

(iii) Why might a decrease in the number of blue tits affect the sparrowhawks more than the stoats?

.....

.....

1 mark

(c) The arrows in the diagram show the direction of energy flow through the food web. A weasel eats a wood mouse. Most of the chemical energy stored in the wood mouse does not end up as chemical energy in the weasel.
Explain why.

.....

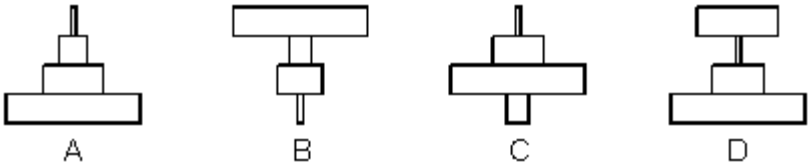
.....

.....

.....

2 marks

(d) The following diagrams show four different **pyramids of numbers**.



A food chain in the food web is

oak trees → **aphids** → **blue tits** → **sparrowhawks**

Which of the drawings, A, B, C or D, best represents the pyramid of numbers for this food chain?

.....

1 mark

Maximum 8 marks

Q1.

(a) any **one** from

- wood mouse
- aphid
- caterpillar
- great tit

2 (L7)

(b) (i) less competition for caterpillars **or** aphids **or** food

*accept 'more food' **or** 'more caterpillars' **or** 'more aphids'*

1 (L7)

(ii) weasels **or** stoats **or** sparrowhawks **or** predators eat more of them

1 (L7)

(iii) any **one** from

- sparrowhawks only eat blue tits and great tits

  accept 'sparrowhawks have more limited diet'

- stoats also eat wood mice

  accept 'stoats eat a greater variety of food'

1 (L7)

(c) any **two** from

- lost as thermal energy **or** by respiration

  *accept 'lost as heat' **or** 'used up by running around **or** hunting **or** working'*

- some material not digested **or** lost in waste
- some material excreted

  accept 'some lost in urine'

  accept 'some material not eaten'

2 (L7)

(d) C

accept a tick or other mark on pyramid C

1 (L7)

[8]